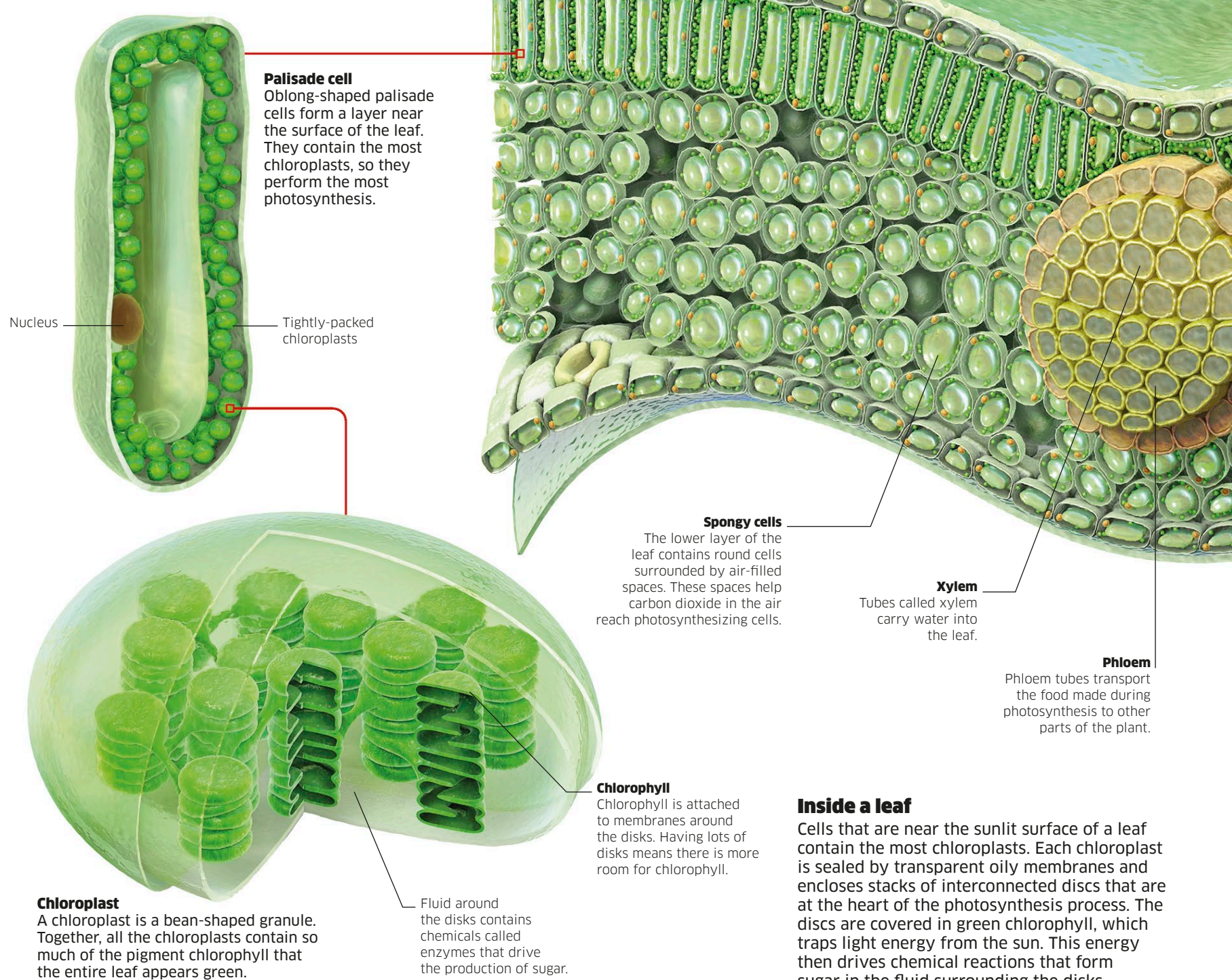


Photosynthesis

Virtually all food chains on Earth begin with photosynthesis—the chemical process in green leaves and algae that is critical for making food.

All around the planet when the sun shines, trillions of microscopic chemical factories called chloroplasts generate enough food to support all the world's vegetation. These vital granules are packed inside the cells of plant leaves and ocean algae. They contain a pigment, called chlorophyll, that makes our planet green and absorbs the sun's energy to change carbon dioxide and water into life-giving sugar.



Inside a leaf

Cells that are near the sunlit surface of a leaf contain the most chloroplasts. Each chloroplast is sealed by transparent oily membranes and encloses stacks of interconnected disks that are at the heart of the photosynthesis process. The disks are covered in green chlorophyll, which traps light energy from the sun. This energy then drives chemical reactions that form sugar in the fluid surrounding the disks.